

Marine Heatwave State Dashboard — User Guide

Sub-arctic to Arctic marine waters around Alaska — Gulf of Alaska, Bering Sea (Eastern and Northern), Chukchi, and Beaufort.

What is a Marine Heatwave?

A **marine heatwave (MHW)** is a prolonged period of unusually warm ocean surface temperatures. Following the Hobday et al. (2016) framework, an MHW is declared when sea surface temperature (SST) exceeds the local 90th-percentile threshold — based on a 1991–2020 baseline — for at least **5 consecutive days**. Brief cool dips of 2 days or fewer are bridged so that a single continuous event is not artificially split.

Once confirmed, each heatwave is tracked with four metrics:

Metric	Symbol	Meaning
Intensity	I	How far SST exceeds the threshold (degrees C)
Duration	D	How many days the event has persisted
Cumulative intensity	C	Total heat exposure over the event (degree-days)
Onset rate	O	How rapidly SST rose at the start of the event (degrees C per day)

These metrics are computed at every 0.25-degree ocean grid cell and then aggregated to regional summaries.

Regions

The dashboard covers five marine regions around Alaska, spanning the sub-arctic North Pacific to the Arctic Ocean:

ID	Region	Latitude	Longitude
GOA	Gulf of Alaska	54–62 N	170–130 W
EBS	Eastern Bering Sea	54–62 N	180–160 W
NBS	Northern Bering Sea	62–67 N	180–160 W
Chukchi	Chukchi Sea	67–73 N	180–155 W
Beaufort	Beaufort Sea	69–73 N	155–130 W

These five regions follow the standard regionalization used by NOAA’s Alaska Fisheries Science Center and the North Pacific Fishery Management Council. Four of them — Gulf of Alaska, East Bering Sea, Chukchi Sea, and Beaufort Sea — are formally recognized as NOAA Large Marine Ecosystems (LMEs). The Northern Bering Sea is treated as its own region (rather than lumped into EBS) because it is a biogeographic transition zone with distinct sea-ice phenology, and has received focused research attention since the 2017–2019 NBS ice failures.

Select any region from the **sidebar dropdown** on either dashboard page. Arctic regions (Chukchi, Beaufort) are ice-masked in winter, so MHW activity is concentrated in the open-water season.

Getting Started

1. **Pick a region** — use the sidebar dropdown on any page to select one of the five regions.
2. **Check current conditions** — the Operational page shows today’s MHW state, event metrics, and risk level.
3. **Explore history** — the Historical page lets you browse 43 years of MHW records (1982–present).

Note: OISST satellite data is typically 1–2 days behind real time, so the most recent date shown may not be today.

Dashboard Pages

Page 1 — Operational

The Operational page shows the **current and recent** MHW state. It has four tabs:

Tab 1: Live MHW Map

A spatial heatmap of the selected region for a given date. Use the date slider to scroll through recent days. Choose from five metrics in the dropdown:

- **Active flag (A)** — binary: is a confirmed MHW in progress at this cell?
- **Intensity (I)** — degrees C above the 90th-percentile threshold
- **Duration (D)** — how many consecutive days the cell has been in MHW
- **Cumulative intensity (C)** — total heat exposure over the event so far (degree-days)
- **Threshold exceedance (x)** — raw SST anomaly above the seasonal threshold (degrees C)

Grid cells under sea ice or on land are masked out. Below the map, summary statistics show the minimum, maximum, and mean values for the selected metric, plus (when viewing Active flag) a count of active cells.

Tab 2: Event Metrics

Time-series charts of five regional aggregate metrics, stacked vertically on a shared time axis:

- **Area fraction** — the share of ocean grid cells in the region experiencing an active MHW (0.25 means 25%)
- **Mean intensity (I)** — average intensity across active cells (degrees C)
- **Mean duration (D)** — average event length across active cells (days)
- **Mean cumulative intensity (C)** — average heat exposure (degree-days)
- **Mean onset rate (O)** — average initial warming speed (degrees C per day)

Use the window dropdown (30, 60, 90, 180 days, **1 year** (default), or Full record) to control the time range. Shaded bands mark periods when area fraction exceeds 5%, indicating a regional-scale event. Below the chart, summary metrics show event days, peak area fraction, and peak date.

Tab 3: Predictability Context

Four stacked subplots on a shared time axis let you visually assess whether large-scale climate regime shifts correspond to changes in heatwave activity:

- **Arctic Oscillation (AO)** — daily bar chart, colored blue (positive) or red (negative). Positive AO tends to confine cold air to the Arctic; negative AO allows cold air outbreaks southward.
- **Pacific Decadal Oscillation (PDO)** — monthly bar chart, colored orange (positive) or blue (negative). Positive PDO is associated with warmer waters in the northeast Pacific.
- **Area fraction** — MHW spatial coverage as a filled line chart, with a dashed line at the 5% regional event threshold.
- **Mean intensity (I)** — average heatwave intensity across active cells (degrees C).

The AO and PDO indices are shown alongside MHW metrics so you can spot correlations — for instance, whether a shift to positive PDO preceded a surge in heatwave coverage. Use the window dropdown (90 days, 180 days, **1 year** (default), or All) to adjust the time range. The “All” view is useful for seeing both PDO phases, since PDO has been in a sustained negative phase since 2020. Below the chart, summary metrics show the latest AO and PDO values, the window-average AO, and the number of MHW event days.

Tab 4: Risk Gauge

A composite risk score (0–100) combining the current values of intensity, duration, cumulative intensity, and onset rate, expressed as percentiles against the full historical record (1982–present). The gauge shows:

- **Overall risk score** — weighted average of the four metric percentiles
- **Individual percentile bars** — how extreme each metric is relative to history
- **30-day trend sparkline** — is risk rising or falling?

Higher risk scores mean current conditions are more extreme than most of the historical record.

Page 2 — Historical (1982–present)

The Historical page lets you explore the full backfill (1982 to the latest available date). A small caption at the top of the page reports the exact date each data source extends through — e.g.,

“MHW data through May 11, 2026 · AO through May 09, 2026 · PDO through Dec 2025” — so “present” always has an explicit meaning.

The page has four tabs:

Tab 1: Annual MHW Burden

A bar chart with one bar per year. Pick which metric the bars represent using the radio buttons above the chart:

- **Peak Area Fraction** — the single highest daily area fraction in that year
- **Mean Area Fraction** — average area fraction across all days in that year
- **Event Days** — count of days where area fraction exceeded 5%

Each bar is colored by where the year ranks in the region’s full historical distribution of the selected metric:

- **Red** — top 10% of years ($\geq$ 90th percentile)
- **Orange** — 50th–90th percentile band
- **Blue** — bottom 50%

The legend above the chart shows the actual numeric cutoffs, which adapt automatically when you change region or metric. The coloring is year-neutral (no hardcoded “Blob years” highlighting): each year gets whatever rank its actual values earn within that region’s history. For some regions the Blob years (2014–2016) come out red on Peak Area Fraction (e.g., GOA and EBS); for others, like Chukchi and Beaufort, the Blob barely registered and those years are blue, while different years (2018–2019 for Chukchi; 2012 for Beaufort) are the extremes.

A small *Pacific Blob* text annotation labels the 2014–2016 period on the chart as a historical-context marker. The label is text-only — it does not affect bar coloring. On regions where the Blob registered as extreme (GOA, EBS), the arrow lands on a tall red bar; on regions where it didn’t (Chukchi, Beaufort), the same arrow lands on a small blue bar, which itself visualizes the regional contrast.

A top-10-years summary table appears below the chart, sorted by Peak Area Fraction.

Tab 2: Event Explorer

Pick a year from the dropdown to drill into that year's MHW activity for the selected region.

- **Five year-level metric cards** at the top: Event Days (days where area fraction > 5%), Peak Area Fraction, Peak Intensity, Peak Duration, Peak Cumulative Intensity.
- **Three stacked time-series subplots** for the chosen year, sharing a common time axis: area fraction (filled crimson line), mean intensity (orange), and mean duration (purple). A dashed red horizontal line marks the 5% regional-event threshold on the area-fraction subplot.
- **Salmon-shaded vertical bands** across all three subplots mark day-spans where area fraction exceeded 5%, so you can visually trace which periods were regional events and how the other metrics behaved during those periods.
- **Monthly summary table** below: per-month Event Days, Peak Area Fraction, Peak Intensity, and Peak Duration.

Tab 3: Metric Distributions

Histograms of daily metric values across the full record, with vertical percentile rulers (25th, 50th, 75th, 90th, 95th). These show how today's conditions compare to the full range of historical variability.

Tab 4: Regime Analysis

Box plots and a summary table comparing MHW metrics across the four AO/PDO regime combinations (AO+ PDO+, AO+ PDO-, AO- PDO+, AO- PDO-). This reveals whether certain climate regimes are associated with more intense or persistent heatwaves.

Important: both the box plots and the median table use event days only — i.e., days where the region's area fraction exceeded 5%. If the panel showed all days (including the ~80% with no MHW activity in any region), every distribution would collapse to zero and the boxes would all look identical. Restricting to event days makes the comparison meaningful.

The summary table reports for each regime:

- **Total days** in that regime
- **Event days** (days where area fraction > 5%)
- **Event rate** (fraction of regime days that were events) — a clean diagnostic for “does this regime host more MHWs?”
- **Medians** of area fraction, intensity, duration, and cumulative intensity over the event days

For Gulf of Alaska, the PDO+ phases (warm Pacific) typically show roughly 2x the event rate of PDO- phases.

Data Sources

Dataset	Provider	Resolution	Coverage
OISST v2.1 (SST + sea ice)	NOAA (NCEI / CoastWatch ERDDAP)	Daily, 0.25 deg	1982–present
Arctic Oscillation	NOAA CPC	Daily	1983–present
Pacific Decadal Oscillation	NOAA PSL	Monthly	1983–present

All data comes from NOAA sources. SST and sea ice are fetched from the same OISST dataset via ERDDAP — cells with ice concentration above 15% are masked out of the MHW analysis. AO is retained from ~1950 and PDO from 1854 so the Regime Analysis tab covers both phases (PDO has been in a sustained negative phase since 2020, so the full record is essential for seeing PDO+ regimes).

In production the dashboard refreshes automatically each day at 14:00 UTC, after OISST publishes new observations (typically by 12:30 UTC). Local development copies are extended via `bash scripts/monthly_refresh.sh` — see the project README.

REST API

A programmatic API is available for researchers who want to access the data directly:

- **Swagger UI:** mhw.iastate.ai/api/docs
- **ReDoc:** mhw.iastate.ai/api/redoc
- **Health check:** mhw.iastate.ai/api/health

All data endpoints are versioned under `/api/v1/`. The service-level health check stays unversioned at `/api/health`.

Endpoint	Description
GET /api/v1/regions	List available regions
GET /api/v1/regions/{region_id}	Single region metadata (date range, row count)
GET /api/v1/regions/{region_id}/states	Daily aggregate time series
GET /api/v1/regions/{region_id}/events	Detected MHW events for a region
GET /api/v1/regions/{region_id}/map?date=...&metric=...	Grid-level state map for a specific date
GET /api/v1/indices/ao	Arctic Oscillation daily values
GET /api/v1/indices/pdo	Pacific Decadal Oscillation monthly values
GET /api/health	Service health (unversioned)

All responses are JSON with snake_case field names. The regional aggregate fields are `area_frac`, `mean_intensity`, `mean_duration`, `cumul_intensity`, and `onset_rate` (internally these correspond to the Hobday-paper symbols I, D, C, O). Categorical fields use typed enums (`MapMetric`, `IndexName`, `IndexFrequency`, `RiskLevel`) so OpenAPI consumers can codegen against them. See the Swagger UI for full parameter details and query options.

Frequently Asked Questions

How often is the data updated? Daily at 14:00 UTC. NOAA typically publishes new OISST data by 12:30 UTC each day.

What does “area fraction” mean? The percentage of ocean grid cells in a region that are currently experiencing a confirmed MHW. An area fraction of 0.25 means 25% of the region is in active heatwave.

Why do Chukchi and Beaufort show zero activity in winter? Those regions are covered by sea ice from roughly November through June. Ice-covered cells are masked out of the MHW analysis because SST under ice is not physically meaningful for heatwave detection.

What is the “Blob”? An exceptionally large and persistent MHW in the northeast Pacific from 2014 to 2016, driven by a persistent atmospheric ridge. It caused widespread marine ecosystem disruption. On the Annual Burden chart, the period is labeled with a *Pacific Blob* text annotation as historical context, but the bar **coloring is not Blob-specific**: it is quantile-based on the displayed metric, so the Blob shows up red automatically on regions where it actually was extreme (GOA, EBS) and stays blue on regions where it wasn't (Chukchi, Beaufort). The label is a calendar marker; the color is the verdict.

Can I download the raw data? Use the REST API endpoints to retrieve daily aggregate and grid-level data in JSON format. The full backfill dataset (1982–present) is available via the API.

What does a risk score of 80 mean? The current MHW conditions are more extreme than roughly 80% of all historical observations (1982–present) for that region.

How is the climatological baseline calculated? For each day of the year at each grid cell, SST values from 1991–2020 within an 11-day centered window are pooled. The mean and 90th percentile of this pool become the climatological reference and detection threshold, respectively.

Why do some metrics show 0 even during warm periods? MHWs require 5 consecutive days above the 90th-percentile threshold before being confirmed. Candidate warm spells of 1–4 days are tracked internally, but all reported metrics (I, D, C, O) remain zero until the 5-day confirmation criterion is met.

What is “gap bridging” and why does a heatwave continue through brief cool dips? If SST dips below the threshold for 2 days or fewer, the event is bridged — treated as a single continuous event rather than split into two separate events. This follows the Hobday et al. (2016) protocol to prevent artificial fragmentation of one physical heatwave.

How is intensity measured — above the mean or above the threshold? In this dashboard, intensity is the anomaly above the **90th-percentile threshold**, not the climatological mean. This is a more conservative definition and yields smaller intensity values than a mean-referenced approach. Keep this distinction in mind when comparing with other MHW studies.

Why do risk scores change even if conditions seem similar? Risk scores are percentile-based against each region's full 1982–present distribution; they are region-specific but not explicitly season-conditioned. A given intensity value may rank as “normal” in the Gulf of Alaska but “extreme” in the Chukchi Sea.

Why are AO and PDO included on the Predictability tab? The Arctic Oscillation and Pacific Decadal Oscillation are large-scale climate modes that influence North Pacific SST patterns. They are shown alongside MHW metrics for regime context — researchers can visually assess whether phase shifts in AO or PDO coincide with changes in heatwave coverage and intensity. They provide correlation context, not direct predictions.

Technical Details

- **Grid resolution:** 0.25 degrees latitude/longitude (OISST native grid)
 - **MHW confirmation:** 5 consecutive days above the 90th-percentile threshold
 - **Gap bridging:** Cool dips of 2 days or fewer are bridged into a single event
 - **Baseline period:** 1991–2020 (30-year climate normal)
 - **Intensity reference:** Anomaly above the 90th-percentile threshold
 - **Regional aggregation:** Area-weighted (cosine of latitude) averages across active cells
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Credits

Marine Heatwave State Dashboard Developed by Rajesh Singh, Professor, Department of Economics, Iowa State University (rsingh@iastate.edu).

Built with Streamlit, FastAPI, and Plotly. Powered by NOAA OISST v2.1, CPC Arctic Oscillation, and PSL Pacific Decadal Oscillation data.

Source code: github.com/rajpython/climate_iastate

Scientific methodology follows: Hobday, A.J. et al. (2016). A hierarchical approach to defining marine heatwaves. *Progress in Oceanography*, 141, 227–238.

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